Assessment of Frailty and Health Outcomes in Patients with COPD; a retrospective study using the Electronic Frailty Index

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Introduction

Why Friliaty Matters?

- Acute exacerbations associated with decline in lung function, worse quality of life, increased mortality
- Aging population: $\sim 16\%$ of adults ≥ 65 are frail
- Potential to predict adverse outcomes independent of age and comorbidity
- **Electronic Frailty Index (eFI)** derived from EHR data

Study Goal: Replicate & extend Pajewski et al. (2019) by assessing the association between frailty (stratified by eFI) and health outcomes/utilization in patients with COPD from their EHR/claims data.

Research Objectives

1. **Describe baseline characteristics** by frailty category
2. **Test predictive ability** of eFI and 36 clinical components
3. **Compare performance** to prior literature

Research Question

Can an eFI computed from routinely collected EHR variables predict hospitalization and mortality?

Electronic Frailty Index: Concept

Deficit Accumulation Theory

- Frailty reflects cumulative health deficits across multiple systems.
- Each deficit adds to overall vulnerability.
- Derived from pre-existing records (ICD codes, labs, meds).
- Predicts hospitalization, mortality, care needs

eFI Calculation

- Quantifies health deficits from EHR data
- 36 clinical deficits counted.
- $\text{eFI} = \frac{\text{deficits present}}{\text{36 total}}$
- Range: 0 (fit) to 1 (most frail).

eFI Categories

Metric	Fit	Pre-Frail	Frail
eFI cutoff	≤ 0.10	0.10–0.21	> 0.21
Our sample	19.4%	47.9%	32.6%

Study Population

Cohort Flow

Retrospective cohort study using EHR data at Atrium Health Wake Forest Baptist

Inclusion Criteria:

- Age ≥55 as of 1/1/2018
- With COPD ≥2 billing codes in prior 2 years (J41, J42, J43, J44)

- Initial $N = 24,929$
- Missing eFI: $6,651$ (26.7%)
- **Analytic sample = 18,278**

Timeframe:

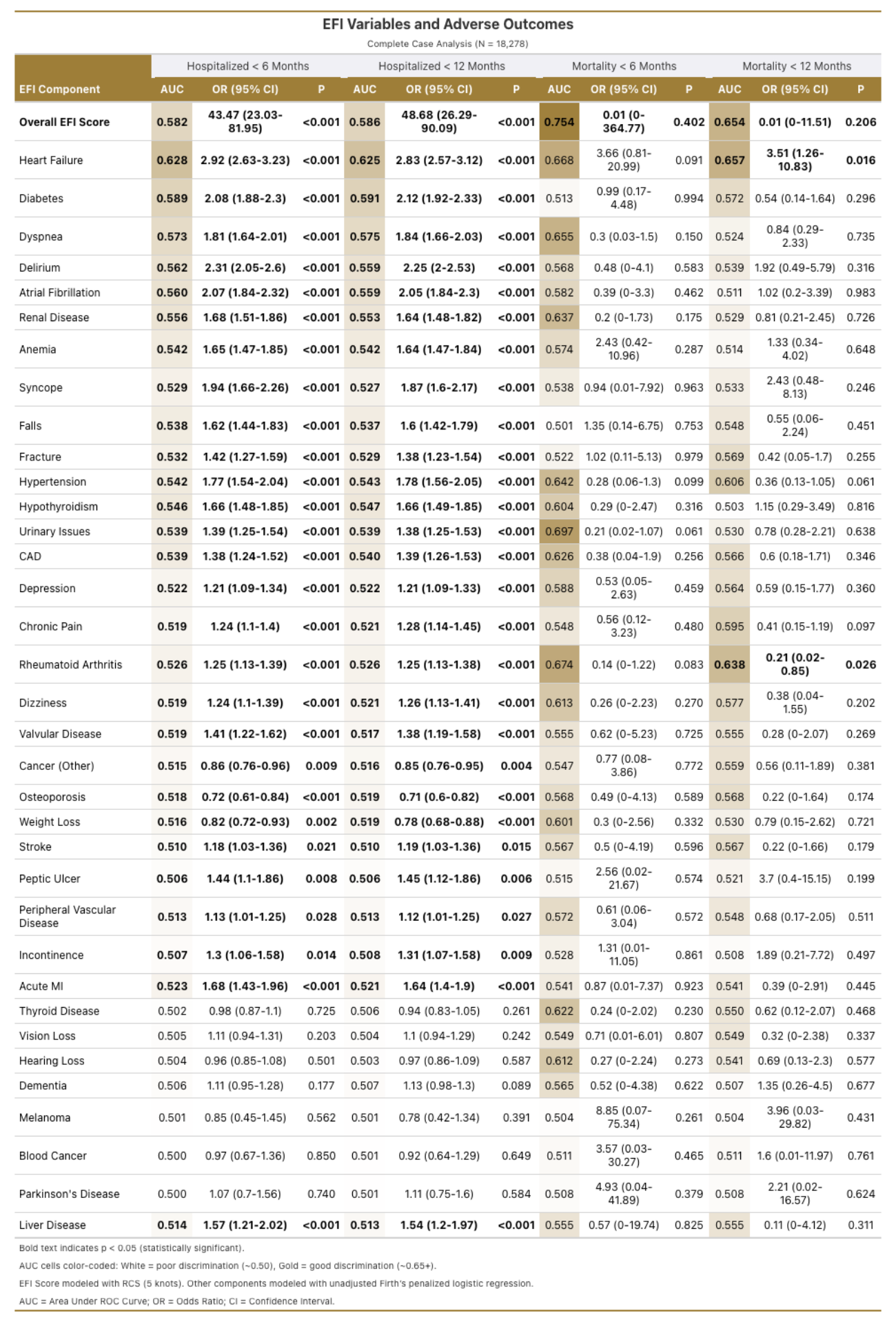
- **eFI calculation:** January 1, 2016 – December 31, 2017
- **Outcome follow-up:** January 1, 2018 – December 31, 2019

Results

The eFI Had Modest Predictive Ability in COPD Patients

Hospitalization AUC = 0.582-0.586 (barely better than random chance at 0.50)

Mortality AUC = 0.754-0.654 but limited events (unstable estimate)



EFI Variables and Adverse Outcomes

Complete Case Analysis (N = 18,278)

	Hospitalized < 6 Months			Hospitalized < 12 Months			Mortality < 6 Months			Mortality < 12 Months		
EFI Component	AUC	OR (95% CI)	P	AUC	OR (95% CI)	P	AUC	OR (95% CI)	P	AUC	OR (95% CI)	P
Overall EFI Score	**0.582**	**43.47 (23.03-81.95)**	**<0.001**	**0.586**	**48.68 (26.29-90.09)**	**<0.001**	**0.754**	**0.01 (0-364.77)**	**0.402**	**0.654**	**0.01 (0-11.51)**	**0.206**
Heart Failure	**0.628**	**2.92 (2.63-3.23)**	**<0.001**	**0.625**	**2.83 (2.57-3.12)**	**<0.001**	0.668	3.66 (0.81-20.99)	0.091	**0.657**	**3.51 (1.26-10.83)**	**0.016**
Diabetes	**0.589**	**2.08 (1.88-2.3)**	**<0.001**	**0.591**	**2.12 (1.92-2.33)**	**<0.001**	0.513	0.99 (0.17-4.48)	0.994	0.572	0.54 (0.14-1.64)	0.296
Dyspnea	**0.573**	**1.81 (1.64-2.01)**	**<0.001**	**0.575**	**1.84 (1.66-2.03)**	**<0.001**	0.655	0.3 (0.03-1.5)	0.150	0.524	0.84 (0.29-2.33)	0.735
Delirium	**0.562**	**2.31 (2.05-2.6)**	**<0.001**	**0.559**	**2.25 (2-2.53)**	**<0.001**	0.568	0.48 (0-4.1)	0.583	0.539	1.92 (0.49-5.79)	0.316
Atrial Fibrillation	**0.560**	**2.07 (1.84-2.32)**	**<0.001**	**0.559**	**2.05 (1.84-2.3)**	**<0.001**	0.582	0.39 (0-3.3)	0.462	0.511	1.02 (0.2-3.39)	0.983
Renal Disease	**0.556**	**1.68 (1.51-1.86)**	**<0.001**	**0.553**	**1.64 (1.48-1.82)**	**<0.001**	0.637	0.2 (0-1.73)	0.175	0.529	0.81 (0.21-2.45)	0.726
Anemia	**0.542**	**1.65 (1.47-1.85)**	**<0.001**	**0.542**	**1.64 (1.47-1.84)**	**<0.001**	0.574	2.43 (0.42-10.96)	0.287	0.514	1.33 (0.34-4.02)	0.648
Syncope	**0.529**	**1.94 (1.66-2.26)**	**<0.001**	**0.527**	**1.87 (1.6-2.17)**	**<0.001**	0.538	0.94 (0.01-7.92)	0.963	0.533	2.43 (0.48-8.13)	0.246
Falls	**0.538**	**1.62 (1.44-1.83)**	**<0.001**	**0.537**	**1.6 (1.42-1.79)**	**<0.001**	0.501	1.35 (0.14-6.75)	0.753	0.548	0.55 (0.06-2.24)	0.451
Fracture	**0.532**	**1.42 (1.27-1.59)**	**<0.001**	**0.529**	**1.38 (1.23-1.54)**	**<0.001**	0.522	1.02 (0.11-5.13)	0.979	0.569	0.42 (0.05-1.7)	0.255
Hypertension	**0.542**	**1.77 (1.54-2.04)**	**<0.001**	**0.543**	**1.78 (1.56-2.05)**	**<0.001**	0.642	0.28 (0.06-1.3)	0.099	0.606	0.36 (0.13-1.05)	0.061
Hypothyroidism	**0.546**	**1.66 (1.48-1.85)**	**<0.001**	**0.547**	**1.66 (1.49-1.85)**	**<0.001**	0.604	0.29 (0-2.47)	0.316	0.503	1.15 (0.29-3.49)	0.816
Urinary Issues	**0.539**	**1.39 (1.25-1.54)**	**<0.001**	**0.539**	**1.38 (1.25-1.53)**	**<0.001**	0.697	0.21 (0.02-1.07)	0.061	0.530	0.78 (0.28-2.21)	0.638
CAD	**0.539**	**1.38 (1.24-1.52)**	**<0.001**	**0.540**	**1.39 (1.26-1.53)**	**<0.001**	0.626	0.38 (0.04-1.9)	0.256	0.566	0.6 (0.18-1.71)	0.346
Depression	**0.522**	**1.21 (1.09-1.34)**	**<0.001**	**0.522**	**1.21 (1.09-1.33)**	**<0.001**	0.588	0.53 (0.05-2.63)	0.459	0.564	0.59 (0.15-1.77)	0.360
Chronic Pain	**0.519**	**1.24 (1.1-1.4)**	**<0.001**	**0.521**	**1.28 (1.14-1.45)**	**<0.001**	0.548	0.56 (0.12-3.23)	0.480	0.595	0.41 (0.15-1.19)	0.097
Rheumatoid Arthritis	**0.526**	**1.25 (1.13-1.39)**	**<0.001**	**0.526**	**1.25 (1.13-1.38)**	**<0.001**	0.674	0.14 (0-1.22)	0.083	**0.638**	**0.21 (0.02-0.85)**	**0.026**
Dizziness	**0.519**	**1.24 (1.1-1.39)**	**<0.001**	**0.521**	**1.26 (1.13-1.41)**	**<0.001**	0.613	0.26 (0-2.23)	0.270	0.577	0.38 (0.04-1.55)	0.202
Valvular Disease	**0.519**	**1.41 (1.22-1.62)**	**<0.001**	**0.517**	**1.38 (1.19-1.58)**	**<0.001**	0.555	0.62 (0-5.23)	0.725	0.555	0.28 (0-2.07)	0.269
Cancer (Other)	**0.515**	**0.86 (0.76-0.96)**	**0.009**	**0.516**	**0.85 (0.76-0.95)**	**0.004**	0.547	0.77 (0.08-3.86)	0.772	0.559	0.56 (0.11-1.89)	0.381
Osteoporosis	**0.518**	**0.72 (0.61-0.84)**	**<0.001**	**0.519**	**0.71 (0.6-0.82)**	**<0.001**	0.568	0.49 (0-4.13)	0.589	0.568	0.22 (0-1.64)	0.174
Weight Loss	**0.516**	**0.82 (0.72-0.93)**	**0.002**	**0.519**	**0.78 (0.68-0.88)**	**<0.001**	0.601	0.3 (0-2.56)	0.332	0.530	0.79 (0.15-2.62)	0.721
Stroke	**0.510**	**1.18 (1.03-1.36)**	**0.021**	**0.510**	**1.19 (1.03-1.36)**	**0.015**	0.567	0.5 (0-4.19)	0.596	0.567	0.22 (0-1.66)	0.179
Peptic Ulcer	**0.506**	**1.44 (1.1-1.86)**	**0.008**	**0.506**	**1.45 (1.12-1.86)**	**0.006**	0.515	2.56 (0.02-21.67)	0.574	0.521	3.7 (0.4-15.15)	0.199
Peripheral Vascular Disease	**0.513**	**1.13 (1.01-1.25)**	**0.028**	**0.513**	**1.12 (1.01-1.25)**	**0.027**	0.572	0.61 (0.06-3.04)	0.572	0.548	0.68 (0.17-2.05)	0.511
Incontinence	**0.507**	**1.3 (1.06-1.58)**	**0.014**	**0.508**	**1.31 (1.07-1.58)**	**0.009**	0.528	1.31 (0.01-11.05)	0.861	0.508	1.89 (0.21-7.72)	0.497
Acute MI	**0.523**	**1.68 (1.43-1.96)**	**<0.001**	**0.521**	**1.64 (1.4-1.9)**	**<0.001**	0.541	0.87 (0.01-7.37)	0.923	0.541	0.39 (0-2.91)	0.445
Thyroid Disease	0.502	0.98 (0.87-1.1)	0.725	0.506	0.94 (0.83-1.05)	0.261	0.622	0.24 (0-2.02)	0.230	0.550	0.62 (0.12-2.07)	0.468
Vision Loss	0.505	1.11 (0.94-1.31)	0.203	0.504	1.1 (0.94-1.29)	0.242	0.549	0.71 (0.01-6.01)	0.807	0.549	0.32 (0-2.38)	0.337
Hearing Loss	0.504	0.96 (0.85-1.08)	0.501	0.503	0.97 (0.86-1.09)	0.587	0.612	0.27 (0-2.24)	0.273	0.541	0.69 (0.13-2.3)	0.577
Dementia	0.506	1.11 (0.95-1.28)	0.177	0.507	1.13 (0.98-1.3)	0.089	0.565	0.52 (0-4.38)	0.622	0.507	1.35 (0.26-4.5)	0.677
Melanoma	0.501	0.85 (0.45-1.45)	0.562	0.501	0.78 (0.42-1.34)	0.391	0.504	8.85 (0.07-75.34)	0.261	0.504	3.96 (0.03-29.82)	0.431
Blood Cancer	0.500	0.97 (0.67-1.36)	0.850	0.501	0.92 (0.64-1.29)	0.649	0.511	3.57 (0.03-30.27)	0.465	0.511	1.6 (0.01-11.97)	0.761
Parkinson's Disease	0.500	1.07 (0.7-1.56)	0.740	0.501	1.11 (0.75-1.6)	0.584	0.508	4.93 (0.04-41.89)	0.379	0.508	2.21 (0.02-16.57)	0.624
Liver Disease	**0.514**	**1.57 (1.21-2.02)**	**<0.001**	**0.513**	**1.54 (1.2-1.97)**	**<0.001**	0.555	0.57 (0-19.74)	0.825	0.555	0.11 (0-4.12)	0.311

Bold text indicates p < 0.05 (statistically significant).
AUC cells color-coded: White = poor discrimination (~0.50), Gold = good discrimination (~0.65+).
EFI Score modeled with RCS (5 knots). Other components modeled with unadjusted Firth's penalized logistic regression.
AUC = Area Under ROC Curve; OR = Odds Ratio; CI = Confidence Interval.

Figure 1. **Discrimination Summary** — Hospitalization AUC [VALUE] indicates **modest predictive performance**; Mortality AUC [VALUE] but limited power due to sparse events.

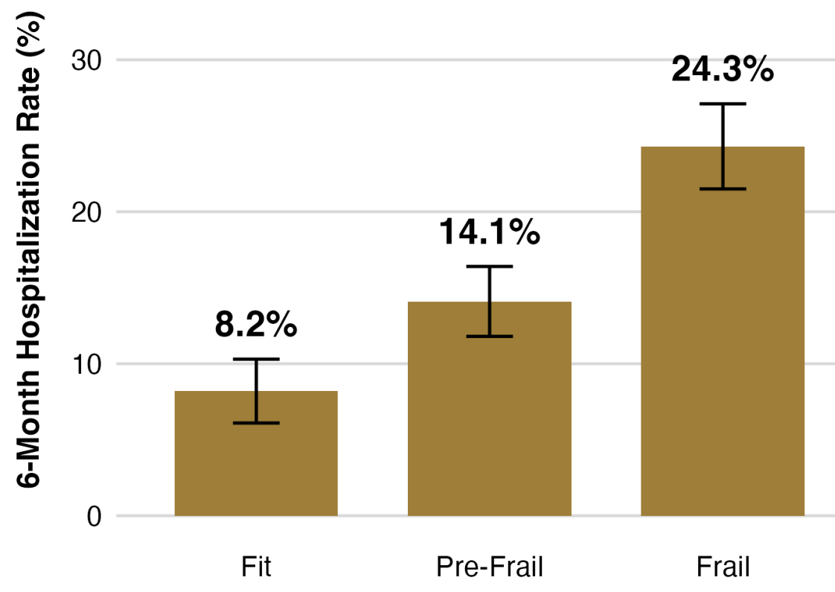


Figure 2. **Healthcare Utilization by Frailty** — Despite modest discrimination, stratification works: [XX]% → [XX]% → [XX]%. Frail vs Fit OR = [VALUE] (95% CI: [LOWER]–[UPPER]).

RCS Predictions with Firth Penalization: Continuous eFI Score and Adverse Outcomes

		Hospitalization				Mortality			
eFI Percentile	eFI Score	6-mo Prob	OR vs 10th	12-mo Prob	OR vs 10th	6-mo Prob	OR vs 10th	12-mo Prob	OR vs 10th
10th	0.070	6.049608%	1.00	6.377143%	1.00	0.3025197%	1.00	0.3195852%	1.00
25th	0.112	7.842594%	1.32	8.358020%	1.34	0.2816990%	0.93	0.2723033%	0.85
50th	0.163	8.891486%	1.52	9.468676%	1.54	0.2387128%	0.79	0.2543891%	0.80
75th	0.218	10.506938%	1.82	11.437263%	1.90	0.2657802%	0.88	0.2093875%	0.65
90th	0.269	12.440430%	2.21	13.511561%	2.29	0.2776121%	0.92	0.2298960%	0.72
95th	0.302	13.813836%	2.49	14.854114%	2.56	0.2744705%	0.91	0.2690442%	0.84

Modeled with restricted cubic splines (5 knots) using Firth's penalized logistic regression.
Probabilities are predicted risks; ORs are relative to the 10th percentile eFI score.

Figure 3. **RCS Predicted Probabilities** — Risk accelerates above 75th percentile; 95th vs 10th percentile ≈ [XX]× odds. However, overall model AUC remains modest.

Outcomes Measured

Primary Outcome:

- **COPD Exacerbations @ 12 months** — Hospital admissions or ER visits with ICD-10 J41–J44

Secondary Outcomes:

- All-cause Mortality (binary)
- Pulmonary Mortality (ICD-10 J41–J44)
- Hospital Admissions (any cause)
- ER Visits (any cause)

Methods

Data Management & Quality Control:

- Cleaned EHR data for 24,929 COPD patients
- Handled missing eFI data; conducted selection bias analysis
- Created analytic dataset of 18,278 patients

Selection Bias Check

Excluded patients were significantly older. Complete-case analysis adopted (no imputation).

Baseline Characteristics by Frailty Category

Mean ± SD or N (%)

Characteristic	Fit	Pre-Frail	Frail	P-value
N	3,554	8,768	5,956	
Age (years)	66.4 ± 6.8	71.2 ± 6.5	76.2 ± 6.1	<0.001
Female (%)	48.2	52.1	57.3	<0.001
CCI Score	2.1 ± 1.8	3.2 ± 2.1	4.8 ± 2.4	<0.001
eFI Score	0.07 ± 0.02	0.15 ± 0.03	0.28 ± 0.06	<0.001

Statistical Modeling:

- Implemented Firth's penalized logistic regression (`logistf`) for rare events
- Applied restricted cubic splines with 5 knots (`rms` package)
- Calculated AUC values and 95% CIs for outcomes
- R packages: `logistf`, `rms`, `dplyr`, `ggplot2`

Data Visualization:

- Created WFU-branded tables using `gt` and `ggplot2`
- Prepared presentation and poster for dissemination

Lessons Learned

About the Field:

- **Intellectual honesty** is essential — reporting modest results transparently builds credibility
- Predictive modeling involves trade-offs between **scalability and accuracy**
- Rare events require specialized methods (e.g., Firth's correction)

Personal & Professional Growth:

- Confidence in independent problem-solving with R
- Translating statistical jargon for clinical audiences
- Reproducible workflows: version control, documentation

Github: https://github.com/xeriksen/predicting_outcomes_using_eFI